

a route through the books already on your shelf

Learning Z

The partition function — what it is, and how to get to it from where you are

An audit of all 144 books in `Pure_Maths_TC_SelfStudy`, a staged route through the ones that matter, and one worked calculation that lands exactly on Part IX.

$$Z = \text{Tr}(T^N)$$

$$\begin{pmatrix} e^{\beta(J+h)} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-h)} \end{pmatrix}$$



$$\lambda_+^N + \lambda_-^N$$

2^N configurations
collapse to two numbers

COVERED

operators, spectral theorem,
trace, measure theory

PARTIAL

probability — two chapters
of Apostol Vol. 2

ABSENT

stat. mech., quantum,
stochastic processes

The finding of the audit. Your shelf has **no** statistical mechanics, **no** quantum mechanics and **no** stochastic processes — Modules 3 through 8 are empty directories.

But it covers the prerequisites unusually well, and one book is filed in the wrong place:

`Linear_Algebra_n_Analysis__by_Kenneth_Kuttler_2020.pdf`, sitting in your linear algebra folder, contains a complete measure theory course in its last five chapters. Route, gaps and a three-week worked calculation follow.

What is in this dossier

- a coverage map of all 144 books
- a staged route, chapter by chapter
- 1D Ising, computed three ways
- the three faces of Z
- the gap list, and what to buy first
- a twelve-week plan

An audit, a route, and one calculation done three ways.

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If you want the short version

§1 tells you what is and is not on your shelf. §3 tells you which four pages of Axler to read first. §5 is the calculation those four pages let you do. §7 is the twelve-week plan.

Two books need buying. Everything else is already in `/Users/zaman/Pure_Maths_TC_SelfStudy`.

1 The audit: what is actually on the shelf

I searched all **144** book-files in /Users/zaman/Pure_Maths_TC_SelfStudy for anything on statistics, thermodynamics, quantum, physics, probability, measure, functional analysis, Hilbert, operators, entropy, stochastic, Markov or Monte Carlo.

The blunt finding

The only matches were a 1993 essay on mathematics-and-theoretical-physics culture, and the two dossiers I wrote you. **There is no statistical mechanics, no quantum mechanics, no probability theory and no stochastic processes on this shelf.** Six of the eight numbered modules are empty directories:

Module_3 DEs & PDEs	Module_4 Diff. Geometry	Module_5 Real Analysis	empty
Module_6 Complex Analysis	Module_7 Stochastic Calculus	Module_8 Math. Statistics	empty

The two in bold are exactly the ones this topic needs.

1.1 Coverage map

	what Z needs	on the shelf?	which book
	operators, spectral theorem	FULL	Axler 2015 §7.B, 7.C, 10.A · Hoffman & Kunze 1971
	inner product spaces	FULL	Axler §6 · Olver & Shakiban ch. 3–4
	eigenvalues, diagonalisation	FULL	Axler §5 · Olver ch. 8–10 · Kuttler ch. 6–7
	multivariable integration	FULL	Apostol Vol. 2 ch. 11 · Spivak, <i>Calculus on Manifolds</i>
	measure & Lebesgue integral	FULL	Kuttler ch. 20–24 (in the LA folder!)
	complex analysis, residues	FULL	Radozycki Part III
	probability	PARTIAL	Apostol Vol. 2 ch. 13–14 only
	Markov chains, MCMC	NONE	— Module_7 is empty
	statistical mechanics	NONE	— nothing on the shelf
	quantum mechanics	NONE	— nothing on the shelf

Figure 1. Coverage of the prerequisites for Z. The line divides what you can study tomorrow from what you would have to acquire.

The find of the audit

Module_1_Linear_Algebra/more-libgen-finds-LA_Books/
Linear_Algebra_n_Analysis__by_Kenneth_Kuttler_2020.pdf
It is filed as a linear algebra book. Its last five chapters are:

Ch.	Title
20	Abstract Measures and Measurable Functions
21	The Abstract Lebesgue Integral
22	Measures From Positive Linear Functionals
23	The L^p Spaces
24	Representation Theorems

That is a complete measure theory course sitting in your linear algebra folder. Chapter 22 is the aptest of all: “measures from positive linear functionals” is the Riesz representation theorem — the statement that *a positive operator determines a measure*. That is Face 1 and Face 3 of Z (next section) meeting in one theorem.

2 What Z is: three faces

$$Z = \int e^{-\beta U(x)} dx \quad (\text{classical}) \qquad Z = \text{Tr}(e^{-\beta \hat{H}}) = \sum_n e^{-\beta E_n} \quad (\text{quantum})$$

These look like different objects. They are one object wearing three hats, and each hat corresponds to a different part of your shelf.

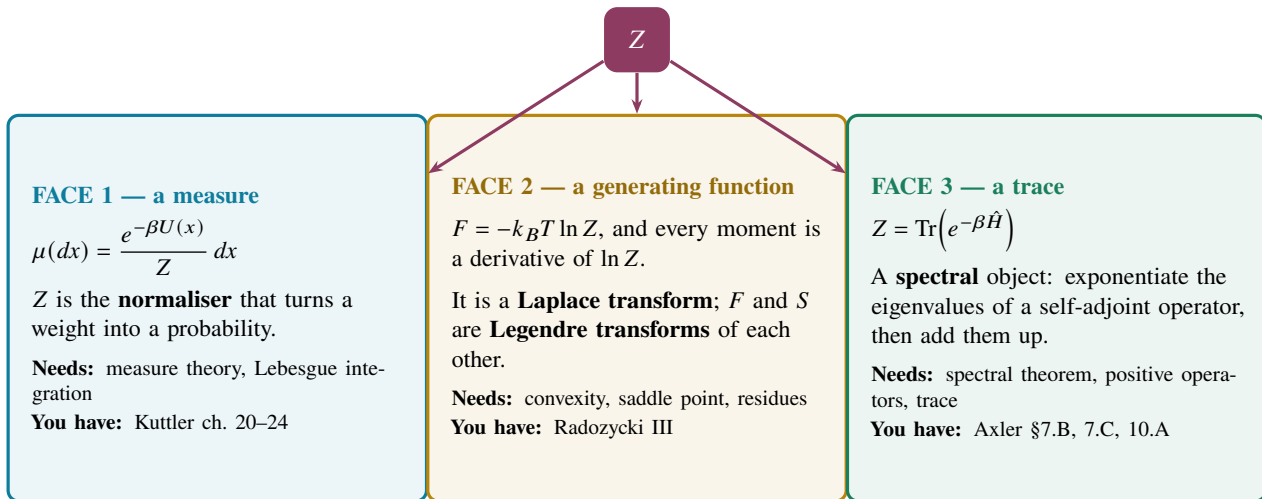


Figure 2. The three faces, each mapped to a book you already own. **Start with Face 3** — it is the one your shelf covers most completely, and it is the one that turns into a concrete calculation fastest.

Why Face 3 first, precisely

$e^{-\beta \hat{H}}$ is not a formal symbol. It is an operator, and it exists because:

1. $\hat{H}$ is **self-adjoint**, so by the **spectral theorem** (Axler §7.B, p. 217 for the complex case) it has an orthonormal eigenbasis with real eigenvalues E_n ;
2. you may therefore *define* $e^{-\beta \hat{H}}$ by exponentiating each eigenvalue;
3. the result is a **positive operator** (Axler §7.C, p. 225) — which is exactly why $Z > 0$ and why μ is a genuine probability measure;
4. and Tr is basis-independent (Axler §10.A, p. 296–299), so $\text{Tr}(e^{-\beta \hat{H}}) = \sum_n e^{-\beta E_n}$ without choosing coordinates.

Four pages of Axler and you have constructed the quantum partition function. No physics book required for this step.

IN PLAIN WORDS

The partition function is one number that summarises every way a system can arrange itself, weighted by how likely each arrangement is. Once you have it, every measurable quantity — energy, entropy, heat capacity, binding affinity — is a derivative of its logarithm. That is why it is called the *partition* function: it tells you how probability is partitioned across states.

3 The route, through books you already own

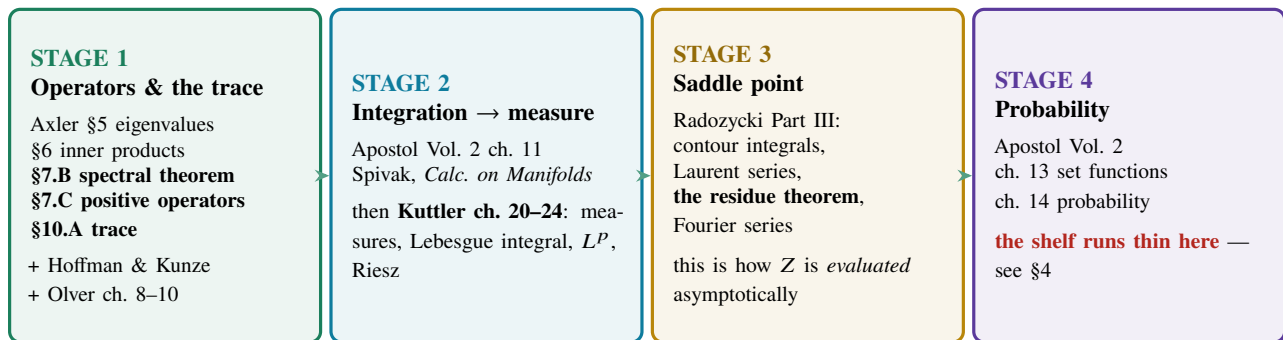


Figure 3. Four stages, all but the last fully covered by books already in the tree.

3.1 Stage 1 in detail — the four pages that matter

I checked the actual contents of your copy of Axler. The three sections you need are:

Axler section	p.	Why $Z = \text{Tr}(e^{-\beta\hat{H}})$ needs it
7.B The Spectral Theorem Complex / Real	217 / 219	This is what lets you <i>define</i> $e^{-\beta\hat{H}}$ at all — you exponentiate the eigenvalues, which requires an orthonormal eigenbasis to exist.
7.C Positive Operators	225	$e^{-\beta\hat{H}}$ is a positive operator. This is why $Z > 0$, and why $e^{-\beta U}/Z$ is a probability density rather than merely a function.
10.A Trace	296–299	“Trace: a connection between operators and matrices” — basis independence is what makes $\text{Tr}(e^{-\beta\hat{H}}) = \sum_n e^{-\beta E_n}$ a definition rather than a coincidence.

You can do Stage 1 this week

Read Axler §7.B, §7.C and §10.A — about 30 pages with the exercises. Then go straight to §5 of this dossier and compute a real partition function with nothing but those three results. That is the whole argument for starting at Face 3.

4 The gaps, and what to buy

Five things are missing. In the order I would fill them:

#	Gap	What I would get	Why this one
1	Statistical mechanics	Chandler , <i>Introduction to Modern Statistical Mechanics</i> (1987) — or Kardar, <i>Statistical Physics of Particles</i> (2007)	If you buy one book, buy Chandler. ~250 pages, written for chemists, reaches the partition function by chapter 3. Kardar is the physicist's book and has the transfer-matrix and renormalisation-group material.
2	Probability, properly	Grimmett & Stirzaker, <i>Probability and Random Processes</i>	Fills Module_8, which is an empty directory.
3	Markov chains & MCMC	Levin & Peres, <i>Markov Chains and Mixing Times</i>	Fills Module_7. This is the book that explains why sampling Z fails — mixing time is the quantitative form of the barrier problem you will meet in §6.
4	Quantum mechanics	Hall , <i>Quantum Theory for Mathematicians</i> — or Sakurai for the physics	Hall is the right one for you: it assumes the mathematics and explains the physics, not the other way round. It picks up exactly where Axler §7 leaves off.
5	Functional analysis	Reed & Simon vol. 1	Only if you pursue the Tier 3 open-quantum-systems problem of Part IX. Kuttler ch. 20–24 covers most of what comes before.

IN PLAIN WORDS

Five books, and only the first is urgent. Everything else on the route you already own. The reason the list is short is that the mathematics you have been building for two years — linear algebra, analysis, measure — *is* the hard part of statistical mechanics. The physics is a thin layer of interpretation on top of it.

5 The worked calculation: 1D Ising, three ways

Everything in this section and the next was computed for this dossier and is reproducible from the companion notebook. It uses numpy only.

5.1 The model, and where the linear algebra enters

N spins $s_i = \pm 1$ on a ring, $E(s) = -J \sum s_i s_{i+1} - h \sum s_i$. Write the Boltzmann weight so each *bond* contributes one factor:

$$e^{-\beta E} = \prod_{i=1}^N \underbrace{\exp\left[\beta J s_i s_{i+1} + \frac{\beta h}{2}(s_i + s_{i+1})\right]}_{T(s_i, s_{i+1})} \quad T = \begin{pmatrix} e^{\beta(J+h)} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-h)} \end{pmatrix}$$

Chaining the factors and closing the ring turns the sum over 2^N configurations into a trace:

$$Z = \sum_{s_1} \cdots \sum_{s_N} T(s_1, s_2) \cdots T(s_N, s_1) = \text{Tr}(T^N) = \lambda_+^N + \lambda_-^N$$

$$\lambda_{\pm} = e^{\beta J} \cosh \beta h \pm \sqrt{e^{2\beta J} \sinh^2 \beta h + e^{-2\beta J}}$$

Notice exactly which theorems did the work

T is symmetric and has positive entries, so **Axler §7.B** gives an orthonormal eigenbasis; **§7.C** says T is a positive operator, so $\lambda_{\pm} > 0$; **§10.A** says the trace is basis-independent, so $\text{Tr}(T^N) = \lambda_+^N + \lambda_-^N$ without choosing coordinates.

A sum over 2^N states collapsed to two numbers, using three sections of a linear algebra textbook you already own.

5.2 Method A versus Method B: they agree to machine precision

N	2^N states	β	Z from $\text{Tr}(T^N)$	Z from brute force
4	16	1.0	1.2123293134×10^2	1.2123293134×10^2
8	256	1.0	9.1604387226×10^3	9.1604387226×10^3
12	4,096	1.0	7.7491438244×10^5	7.7491438244×10^5
16	65,536	0.2	9.0061138775×10^4	9.0061138775×10^4
16	65,536	0.5	4.4784000941×10^5	4.4784000941×10^5
16	65,536	1.0	6.8584536591×10^7	6.8584536591×10^7

Table 1. Worst relative disagreement across all twelve test cases: 7.56×10^{-13} — machine precision. A sum over 65,536 configurations and the trace of a 2×2 matrix are the same number.

5.3 Thermodynamics: everything is a derivative of $\ln Z$

As $N \rightarrow \infty$ only the larger eigenvalue survives, because $(\lambda_-/\lambda_+)^N \rightarrow 0$. At $h = 0$, $\lambda_+ = 2 \cosh \beta J$, so

$$f = -\frac{1}{\beta} \ln(2 \cosh \beta J), \quad u = -J \tanh \beta J, \quad \frac{C}{k_B} = \left(\frac{\beta J}{\cosh \beta J} \right)^2$$

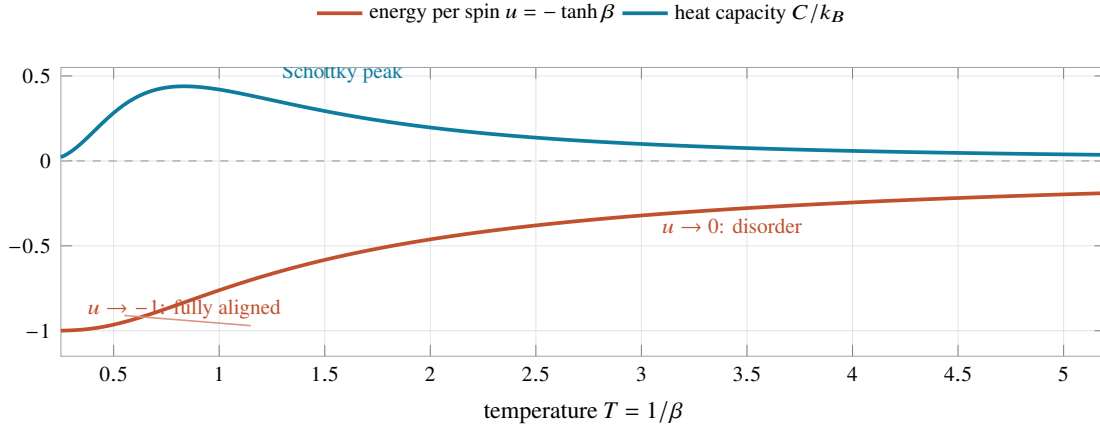


Figure 4. Exact thermodynamics of the 1D Ising chain, from $\ln \lambda_+$ alone. Note there is *no* phase transition: f is smooth for all $T > 0$. That is a theorem — a one-dimensional system with short-range interactions cannot order at finite temperature. The bump in C is a Schottky anomaly, not a critical point.

5.4 The eigenvalue gap is the correlation length

This is the deepest line in the whole calculation, and it is one line of linear algebra:

$$\langle s_0 s_r \rangle = \left(\frac{\lambda_-}{\lambda_+} \right)^r = e^{-r/\xi}, \quad \xi = \frac{1}{\ln(\lambda_+/\lambda_-)}$$

β	λ_+	λ_-	λ_+/λ_-	ξ
0.2	2.04013	0.40267	5.0665	0.616
0.5	2.25525	1.04219	2.1640	1.295
1.0	3.08616	2.35040	1.3130	3.672
1.5	4.70482	4.25856	1.1048	10.034
2.0	7.52439	7.25372	1.0373	27.296
2.5	12.26458	12.10041	1.0136	74.205
3.0	20.13532	20.03575	1.0050	201.714

Table 2. As temperature falls the spectral gap closes and the correlation length grows *exponentially* — from 0.6 to 202 lattice spacings. **Keep this sentence: the spectral gap of the transfer matrix controls how far correlations reach.** It returns in §6 as the reason Monte Carlo fails, and in Part IX §6 as the reason the transfer *operator* is the right object for molecular kinetics.

6 Where Monte Carlo breaks — measured

Now the method everyone actually uses on systems that are *not* exactly solvable: Metropolis. Propose a spin flip, accept with probability $\min(1, e^{-\beta\Delta E})$, run long enough.

That last phrase is where everything goes wrong, and the Ising model lets us measure exactly how wrong.

A perfect test, because we know the answer exactly

At $h = 0$ the model is exactly symmetric under flipping every spin. Therefore

$$\langle M \rangle = 0 \quad \text{exactly, for every } \beta, \text{ every } N.$$

No approximation, no limit. So: run several *independent* chains and see whether they all return zero. **Every chain below runs the identical, correct algorithm. None of them is buggy.**

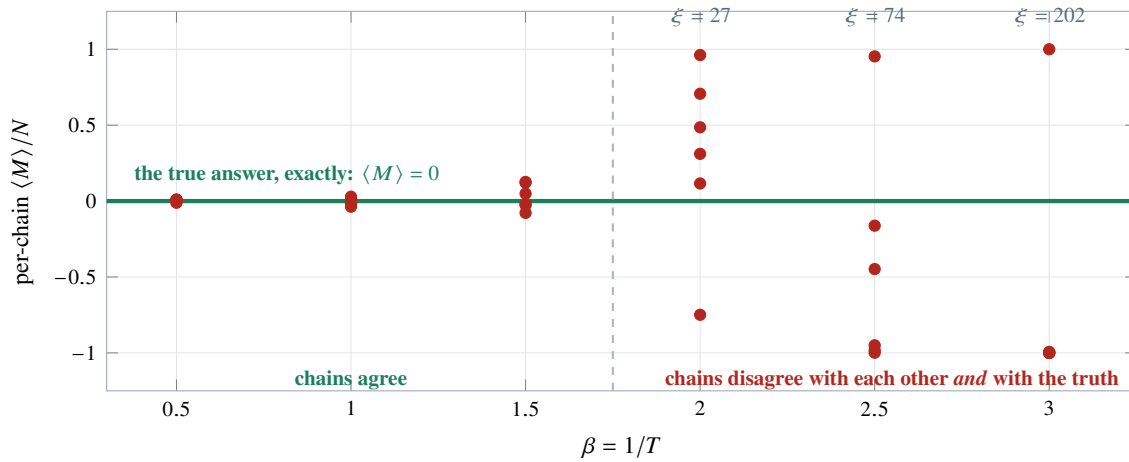


Figure 5. Six independent Metropolis chains, $N = 64$, 6,000 sweeps each. Measured for this dossier. At $\beta = 3$ the chains report ± 1 when the exact answer is 0.

β	ξ	spread across chains	verdict
0.5	1.30	0.021	ok
1.0	3.67	0.066	ok
1.5	10.03	0.203	poor
2.0	27.30	1.710	BROKEN
2.5	74.21	1.952	BROKEN
3.0	201.71	2.000	BROKEN

Table 3. A spread of 2.000 means some chains returned +1 and others -1 — the maximum possible disagreement. An independent 8-chain run at 30,000 sweeps reproduced the same pattern.

Why — and it is the same mechanism as the whole of Part IX

As β grows, ξ grows exponentially (Table 2). Once $\xi \gtrsim N$ the entire ring is a single domain, and to get from “mostly up” to “mostly down” the chain must pass through a high-energy configuration. The waiting time for that is $\sim e^{\beta\Delta E}$ — **exponential in the barrier height**.



This is the quantitative form of the sentence in Part IX §2: “*Monte Carlo converges at $O(n^{-1/2})$ provided it mixes. On a rugged landscape with barriers $\gg k_B T$, it does not mix.*” And it is the argument for Boltzmann generators: rather than *crossing* the barrier, learn a change of variables in which there is no barrier to cross.

**IN
PLAIN
WORDS**

The lesson is not “Monte Carlo is bad”. It is that a method can be provably correct, correctly implemented, and still give you a confidently wrong answer — because “run it long enough” hid an exponential. You would never know, in a system where the true answer was not available. That is the situation in every protein simulation ever run.

7 The bridge to Part IX

Everything in §5 and §6 generalises, and the generalisation is exactly what Part IX was about.

Here — 1D Ising	There — molecular systems (Part IX)
T , a 2×2 transfer matrix	$\mathcal{P}_t$, the transfer operator on conformation space
$Z = \text{Tr}(T^N)$, a sum over 2^N states	$Z = \int e^{-\beta U(x)} dx$, uncountably many states
largest eigenvalue $\lambda_+ \Rightarrow$ free energy	leading eigenvector $\Rightarrow$ the equilibrium measure μ
gap $\lambda_+/\lambda_- \Rightarrow$ correlation length ξ	remaining eigenvalues $\Rightarrow$ relaxation timescales $t_i = -\tau/\ln \lambda_i$
eigenvectors of a 2×2 matrix	eigenvectors $\Rightarrow$ the slow collective variables (what VAMPnets learn)
MCMC fails when $\xi \gtrsim N$	MCMC fails when barriers $\gg k_B T$ — the same mechanism

The sentence to keep

It is called a transfer operator because it is the continuum limit of the transfer matrix you just diagonalised.

When Part IX said “the leading eigenvector is the equilibrium measure, and the remaining eigenvalues give the relaxation timescales”, that was not an analogy. It is literally Table 2 of this dossier, with 2×2 replaced by infinite-dimensional.

7.1 What the Ising route does *not* teach you

Being straight, so you do not over-learn from one example:

What it does not show	Where to get it
Why computing Z is #P-hard. In 1D it is not hard — the transfer matrix is exact.	2D (Onsager 1944) and higher. Kardar has the transfer-matrix route into 2D.
Phase transitions. 1D short-range systems cannot order at $T > 0$.	Chandler ch. 5; Kardar on the Ising model in $d \geq 2$ and the renormalisation group.
Continuous configuration space. Here the state space is finite.	Chandler on classical fluids; then Part IX §4 on configurational entropy.

8 A twelve-week plan

Weeks	Read	Do	Question to answer in writing
1–2	Axler §5, §6, §7.B, §7.C, §10.A	Prove $\text{Tr}(T^N) = \lambda_+^N + \lambda_-^N$ from basis-independence of the trace	Why does $e^{-\beta\hat{H}}$ need the spectral theorem to even be defined?
3	—	Run the companion notebook. Do exercises 1–3	At what N does brute force become impossible on your laptop? Extrapolate to $N = 60$.
4–5	Chandler ch. 1–3 (the one book to buy)	Re-derive f , u , C for the Ising chain without looking	Which of the six quantities in Part IX Fig. 2 can you now compute yourself?
6–7	Radozycki III: contour integrals, residues	Evaluate a Gaussian integral by contour deformation; then by saddle point	What <i>is</i> the saddle-point approximation approximating?
8–9	Kuttler ch. 20–21	Write down $\mu(dx) = e^{-\beta U} dx / Z$ as a measure and verify it is a probability measure	Where exactly does positivity of $e^{-\beta\hat{H}}$ get used?
10	Kuttler ch. 22 (Riesz)	—	In what sense does a positive operator <i>determine</i> a measure?
11–12	Levin & Peres ch. 1–4 (the second book to buy)	Notebook exercise 4: measure autocorrelation time vs ξ	Restate §6 of this dossier in the language of mixing times.

Table 4. Two books purchased, ten weeks of reading you already own. At the end of week 12 you can read Part IX §7 (Boltzmann generators) and know precisely what problem it is solving.

The companion notebook

Partition_Function_Study/Partition_Function_01_Ising_Transfer_Matrix.ipynb

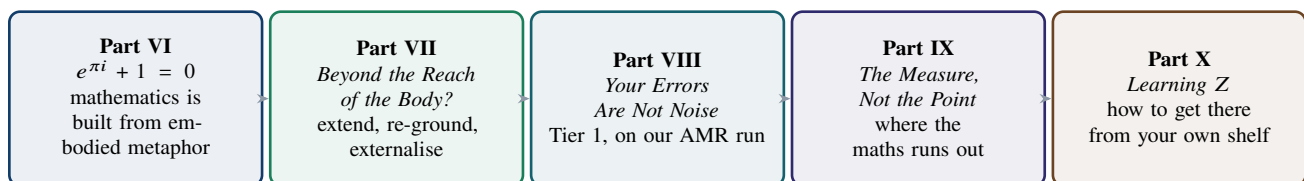
17 cells — 9 exposition, 8 code. Runs end to end on `numpy` alone; plots are optional and it degrades gracefully if `matplotlib` is absent. Every cell was executed before delivery.

§	Contents
1–2	the model; deriving the transfer matrix from the Boltzmann weight
3	Method B — brute force over all 2^N states, and the agreement table
4	thermodynamics: f , u , C from $\ln \lambda_+$
5	the eigenvalue gap and the correlation length
6–7	Method C — Metropolis, and the ergodicity failure test
8	the bridge table to Part IX
9	six exercises, including “find the 2^N wall” and the 2D transfer matrix

9 Where everything came from

Claim	Source
From your own shelf — verified by opening the files	
144 book-files; Modules 3–8 empty	find over /Users/zaman/Pure_Maths_TC_SelfStudy
Axler §7.B p. 217/219, §7.C p. 225, §10.A p. 296	Module_1_Linear_Algebra/Tensor-College-Referred-Books/Linear_Algebra_Done_Right__by_Axler_2015.pdf
Kuttler ch. 20–24: measures, Lebesgue, L^p , Riesz	Module_1_Linear_Algebra/more-libgen-finds-LA_Books/Linear_Algebra_n_Analysis__by_Kenneth_Kuttler_2020.pdf
Apostol Vol. 2 ch. 13 “Set Functions and Elementary Probability”	Module_2_Multivariable_Calculus/Tom_M.Apostol__Calculus_Vol2_MVC_2dEd_197
Radozycki III: contour integrals, residues, Fourier series	Foundation_Math_Books_z/Solving_Problems_in_Mathematical_Analysis_Part_II
Olver & Shakiban ch. 8–10 eigenvalues, incl. infinite dimensions	Module_1_Linear_Algebra/more-libgen-finds-LA_Books/Applied_Linear_Algebra
Computed for this dossier	
Tables 1–3, Figures 4–5	companion notebook; numpy 1.26.4, seeds fixed, fully reproducible
Worst A-vs-B disagreement 7.56×10^{-13}	12 test cases, $N \in \{4, 8, 12, 16\}$, $\beta \in \{0.2, 0.5, 1.0\}$
Ergodicity failure	6 chains $\times$ 6,000 sweeps; confirmed by an independent 8-chain, 30,000-sweep run
Recommended acquisitions	
Chandler, <i>Introduction to Modern Statistical Mechanics</i> (1987)	the one book to buy first
Levin & Peres, <i>Markov Chains and Mixing Times</i>	why sampling Z fails, quantitatively
Grimmett & Stirzaker, <i>Probability and Random Processes</i>	fills Module_8
Hall, <i>Quantum Theory for Mathematicians</i>	picks up where Axler §7 stops
Kardar, <i>Statistical Physics of Particles</i> (2007)	transfer matrices in 2D, renormalisation group

9.1 How this dossier relates to the others



2^N configurations collapse to two numbers.

That is the partition function in one sentence, and three sections of a linear algebra book you already own are enough to prove it. Everything after that — measures, rates, why sampling fails, why Boltzmann generators exist — is the same idea in more dimensions.